BEYOND 40°C

QUANTIFYING THE HUMID-HEAT BLIND SPOT IN INDIA'S HEATWAVE WARNINGS

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[BUILT END-TO-END IN R]

PROBLEM STATEMENT

India declares a heatwave only when the dry-bulb temperature crosses a fixed line: 40 °C on the plains or 37 °C on the coast, plus a 4.5 °C departure from normal. That declaration is what switches on city and state Heat Action Plans.

But the body cools itself by evaporating sweat, and humid air shuts that down. A 36 °C day at 60% humidity has a wet-bulb temperature of about 29.4 °C, close to the limit at which healthy young adults can no longer stabilise their core temperature. Yet it never triggers an alert.

A dependable, probabilistic way to detect and forecast this humid heat is needed.

LITERATURE REVIEW

Official criteria. IMD declares a heatwave when a station reaches 40 °C (plains), 37 °C (coast) or 30 °C (hills) with a departure of at least 4.5 °C from normal, on two stations for two consecutive days. IMD's experimental Heat Index (NOAA equation; green < 35, yellow 36–45, orange 46–55, red > 55 °C) is guidance only and is not part of the declaration.

Physiology. Vecellio et al. (2022, J. Appl. Physiol.) measured critical wet-bulb limits of about 30–31 °C for young, healthy adults in warm-humid air, well below the theoretical 35 °C. Raymond et al. (2020, Science Advances) report that extreme humid heat has more than doubled in frequency since 1979.

Policy. A Centre for Policy Research review (2023) of 37 Heat Action Plans across 18 states found that only 10 set local thresholds. Plans focus on dry heat and largely ignore humid heat and warm nights.

Statistics. Extreme-value theory (Coles, 2001), Mann–Kendall/Sen trend tests and probabilistic verification (Brier, ROC) are standard climate tools. To our knowledge, they have not been combined into a day-by-day audit of the Indian rule against a physiological humid-heat criterion.

The Gap

Dry-bulb only trigger: 40 °C plains, 37 °C coast

Humid heat and warm nights ignored by most plans

Only 10 of 37 Heat Action Plans use local thresholds

No probabilities, return periods or uncertainty

How does our project close it?

- ↗ Audits the official rule day by day, 10 cities × 45 years
- ↗ Judges danger by physiology (wet-bulb), not dry-bulb
- ↗ Quantifies uncertainty: CIs, posteriors, bootstrap
- ↗ Tests whether the blind spot is growing (trends, EVT)
- ↗ Ships a calibrated next-day warning + dashboard, in R

OUR PROJECT

RQ1 What share of dangerous humid-heat days does the IMD rule miss?

Interval estimation · χ^2, McNemar, κ · Bayesian posterior

RQ2 Are danger days becoming more frequent?

Mann–Kendall · negative-binomial GLM · ANOVA · permutation

RQ3 How extreme can humid heat get, and is the tail moving?

GEV maximum likelihood · return levels · likelihood-ratio test

RQ4 Do dangerous days cluster into spells?

Two-state Markov chain · geometric law · Monte Carlo

RQ5 Can we warn a day ahead with honest probabilities?

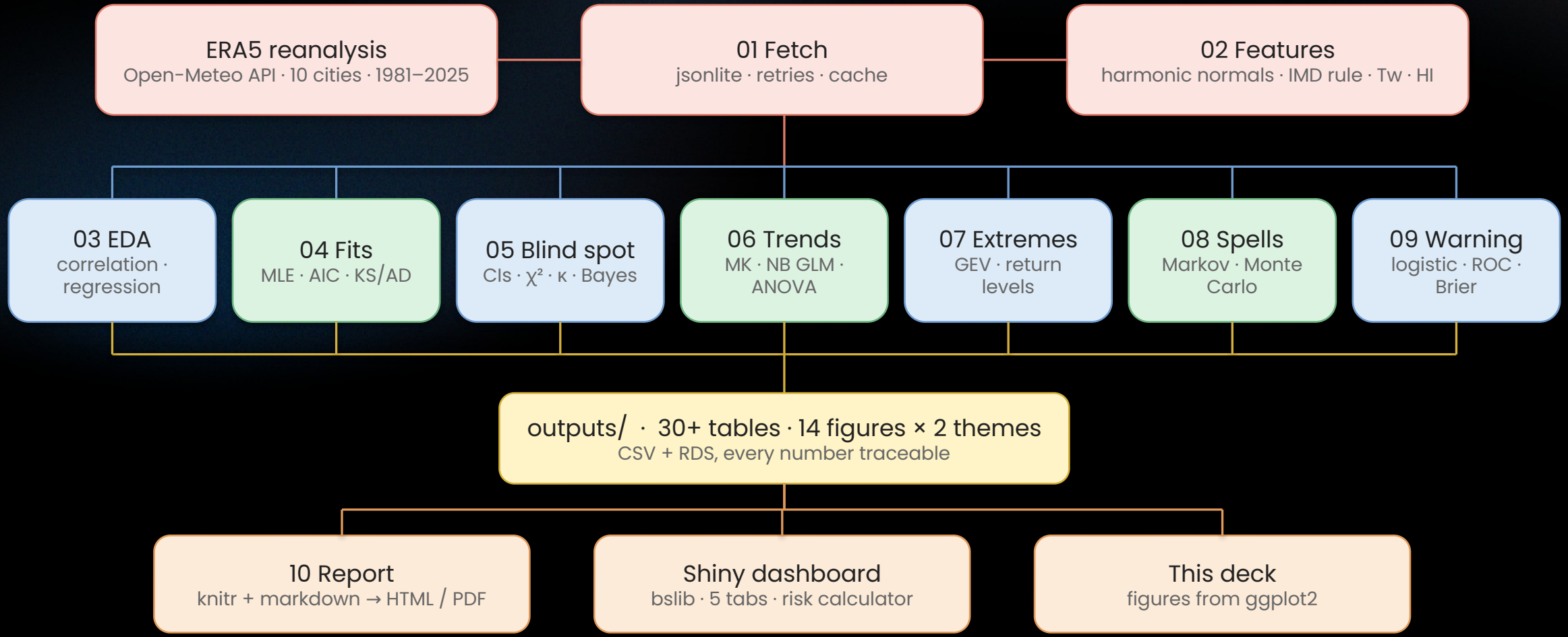
Logistic regression · Brier skill · ROC/AUC · reliability

RQ6 Can baseline ML warn 1–3 days ahead, even in a city it never saw?

Decision tree · random forest · hybrid IMD + ML alert · leave-one-city-out

DATA, METHODS & BLOCK DIAGRAM

PIPELINE BLOCK DIAGRAM



Rscript run_all.R reproduces everything from raw data in under a minute.

DATA

164,360
city-days, no missing values

10
cities: 7 plains, 3 coastal

45
years, 1981–2025 (ERA5)

6
daily variables per city

Plains rule (≥ 40 °C): Delhi, Lucknow, Ahmedabad, Nagpur, Kolkata, Bhubaneswar, Vellore

Coastal rule (≥ 37 °C): Chennai, Mumbai, Visakhapatnam

Variables: Tmax, Tmin, daily max and mean wet-bulb, RH min and mean

Danger day: daily max wet-bulb ≥ 29 °C

Critical day: wet-bulb ≥ 31 °C (Vecellio 2022)

IMD heatwave day: rule met on ≥ 2 consecutive days, departure from a harmonic-regression normal

Physics check: Stull wet-bulb vs ERA5, r = 0.96–0.99

HOW R POWERS THE PROJECT

WRITTEN FROM SCRATCH

GEV log-likelihood + MLE (optim, Hessian)

Return levels: delta method + bootstrap

Mann–Kendall test + Sen's slope

Wilson CI · Cohen's κ · Beta posterior

Year-block bootstrap · permutation test

Markov chain + Monte Carlo simulator

IMD rule engine · Stull wet-bulb · NWS HI

BASE R STATISTICS

lm · glm (binomial, Poisson) · aov

TukeyHSD · friedman.test · kruskal.test

chisq · fisher · mcnemar · prop · binom

t.test · wilcox.test · ks.test

shapiro.test · bartlett.test · cor.test

optim · optimHess · uniroot · rle

PACKAGES

dplyr · tidyr: wrangling 164k rows

ggplot2 · patchwork · ggridges: 14 figures

fitdistrplus · MASS::glm.nb · evd (check)

pROC: ROC + DeLong intervals

jsonlite: API download

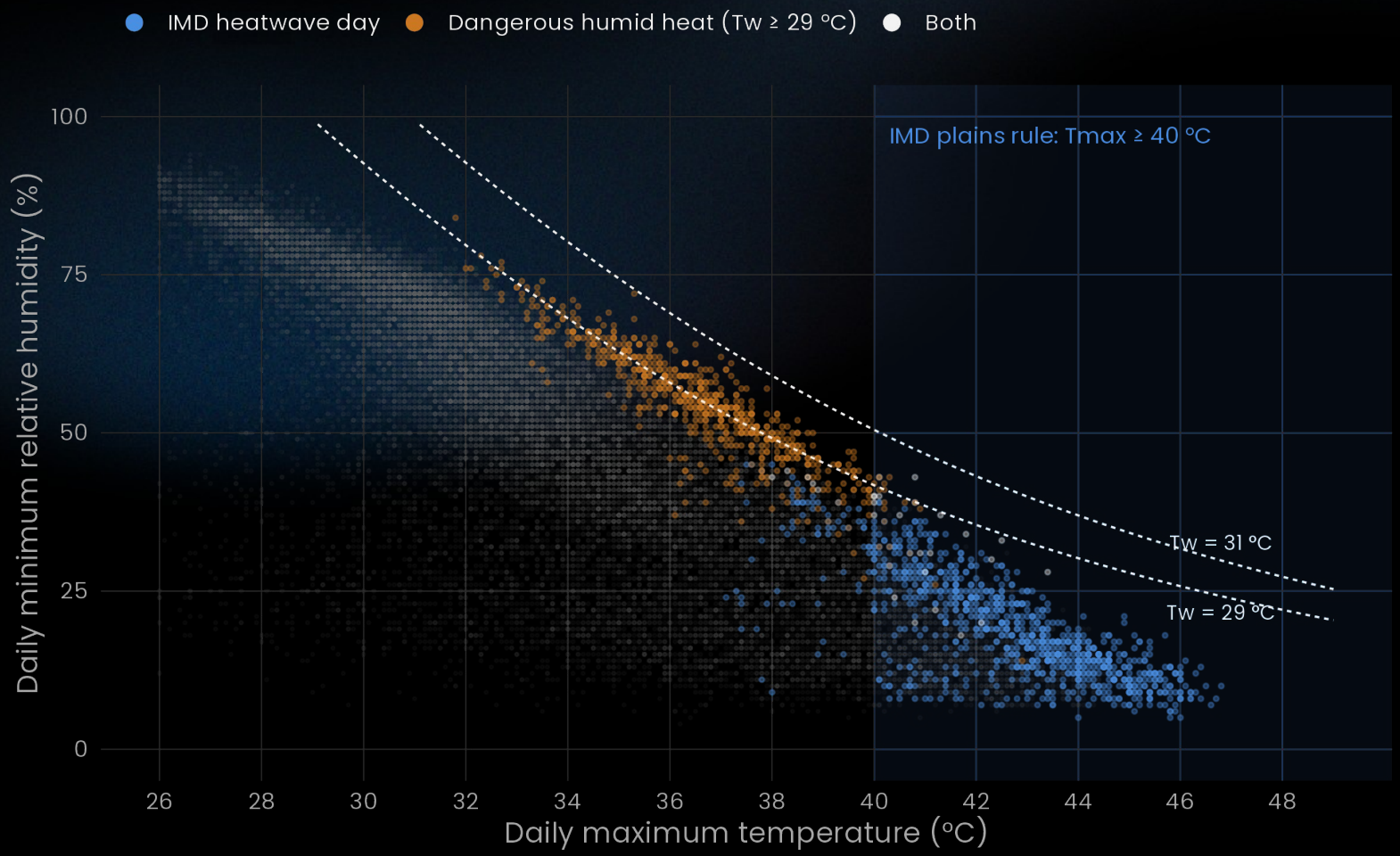
shiny · bslib: dashboard

knitr · markdown: report

≈ 1,700 lines of R · 10 modules · 1 command · our MLE matches evd::fgev to every printed digit

RESULTS

WHY A DRY-BULB RULE CANNOT SEE HUMID HEAT



10 cities, Mar–Sep 1981–2025, ERA5. Dashed lines: wet-bulb isotherms (Stull 2011).

r = -0.61 to -0.91

Tmax vs RH in every city ($p < 0.001$): the hottest days are the driest

+1.5–2.1 °C

wet-bulb rise per +10% humidity at the same temperature (regression)

27 of 990

IMD heatwave days were also humid-danger days

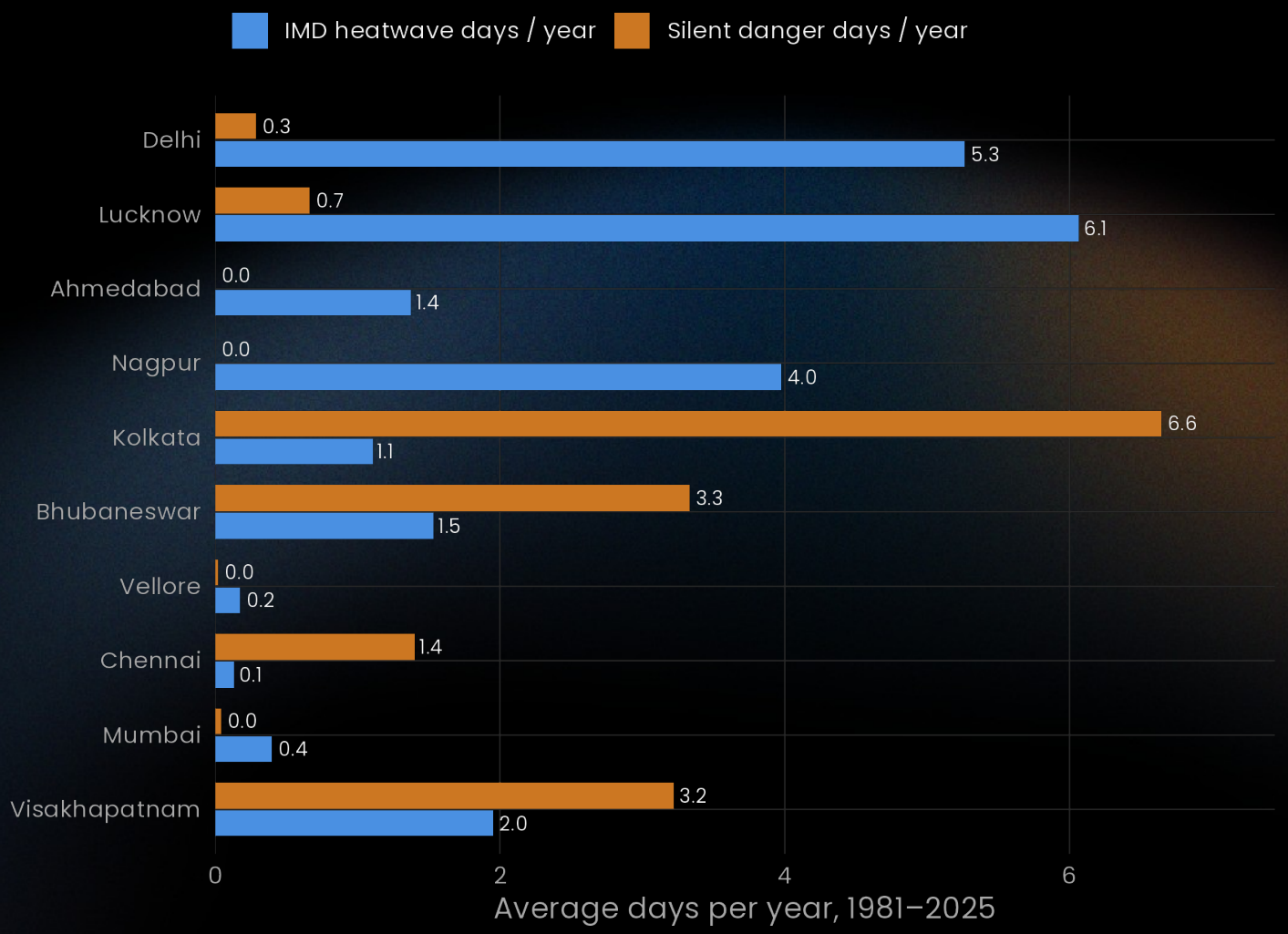
RQ1 · THE BLIND SPOT

96.3%

of dangerous humid-heat days (703 of 730) were never IMD heatwave days

Wilson 95% CI 94.7%–97.4%

Our rule is more generous than IMD's, so this is a lower bound



RQ1 · IS IT REAL? THE STATISTICAL EVIDENCE

Cohen's κ ≈ 0 (at most 0.09): no agreement between the two definitions

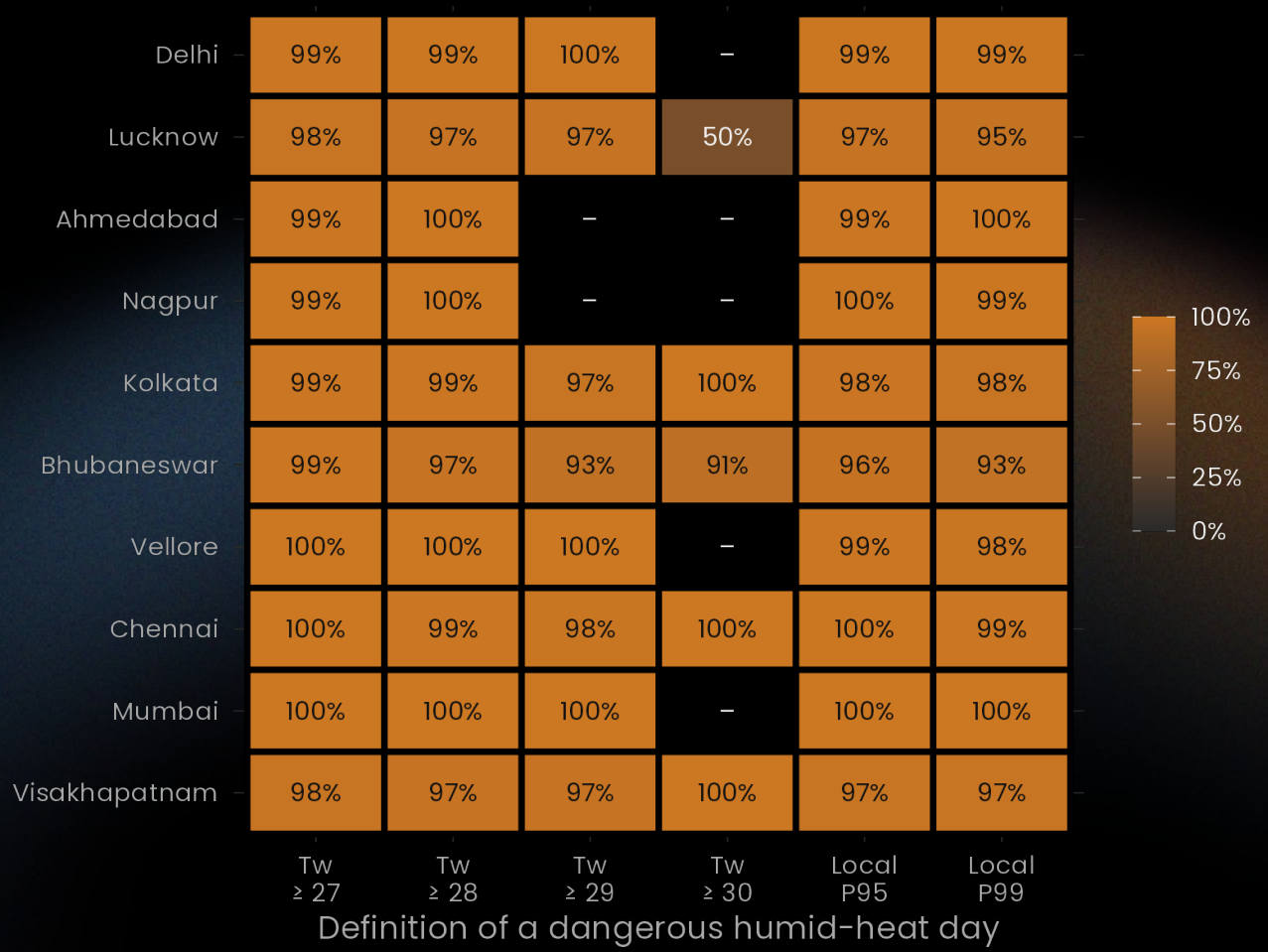
McNemar $p < 0.05$ in every city: the disagreement is systematic

χ^2 homogeneity $p = 0.33$: same miss rate in every city

Coastal vs plains $p = 0.52$: a property of the rule, not the region

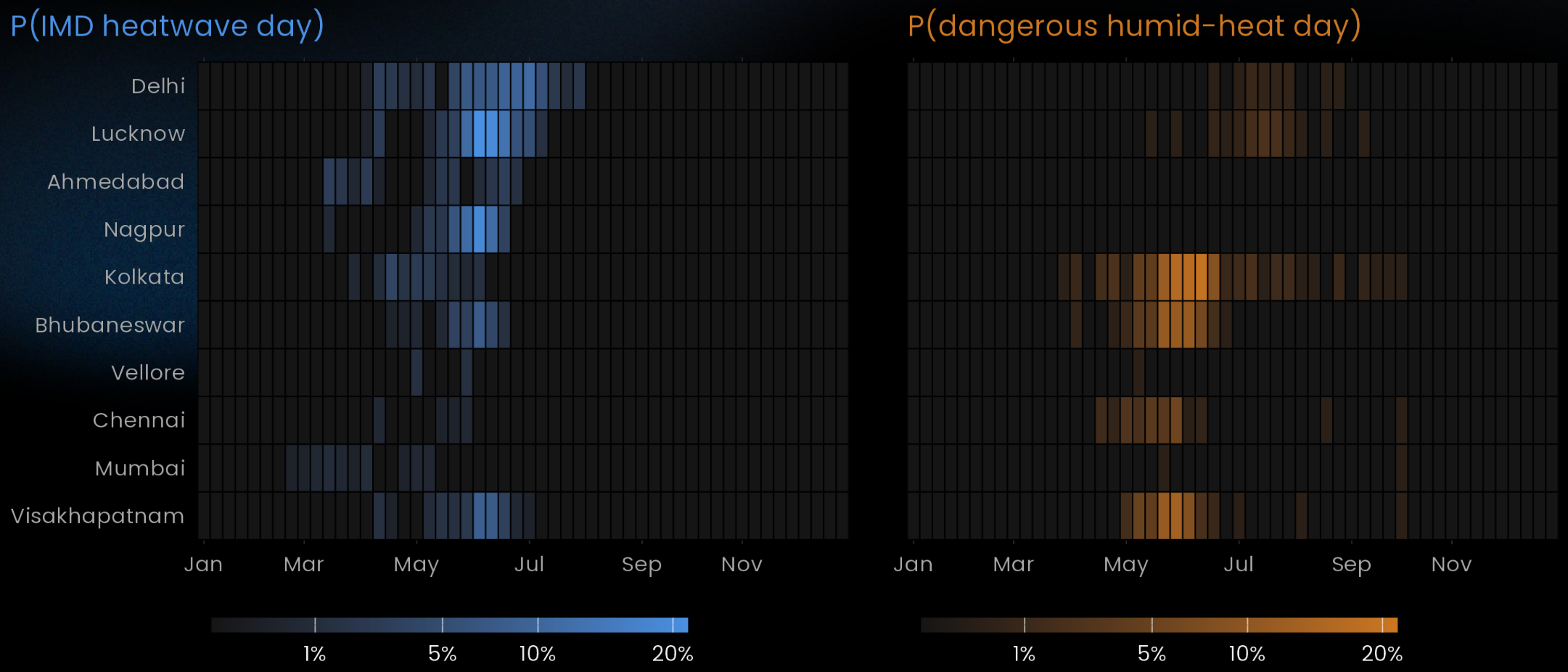
Bayesian Beta(1,1) prior: posterior miss rate > 88% in every city

Heat Index IMD's own experimental index was orange/red on 81% of these days



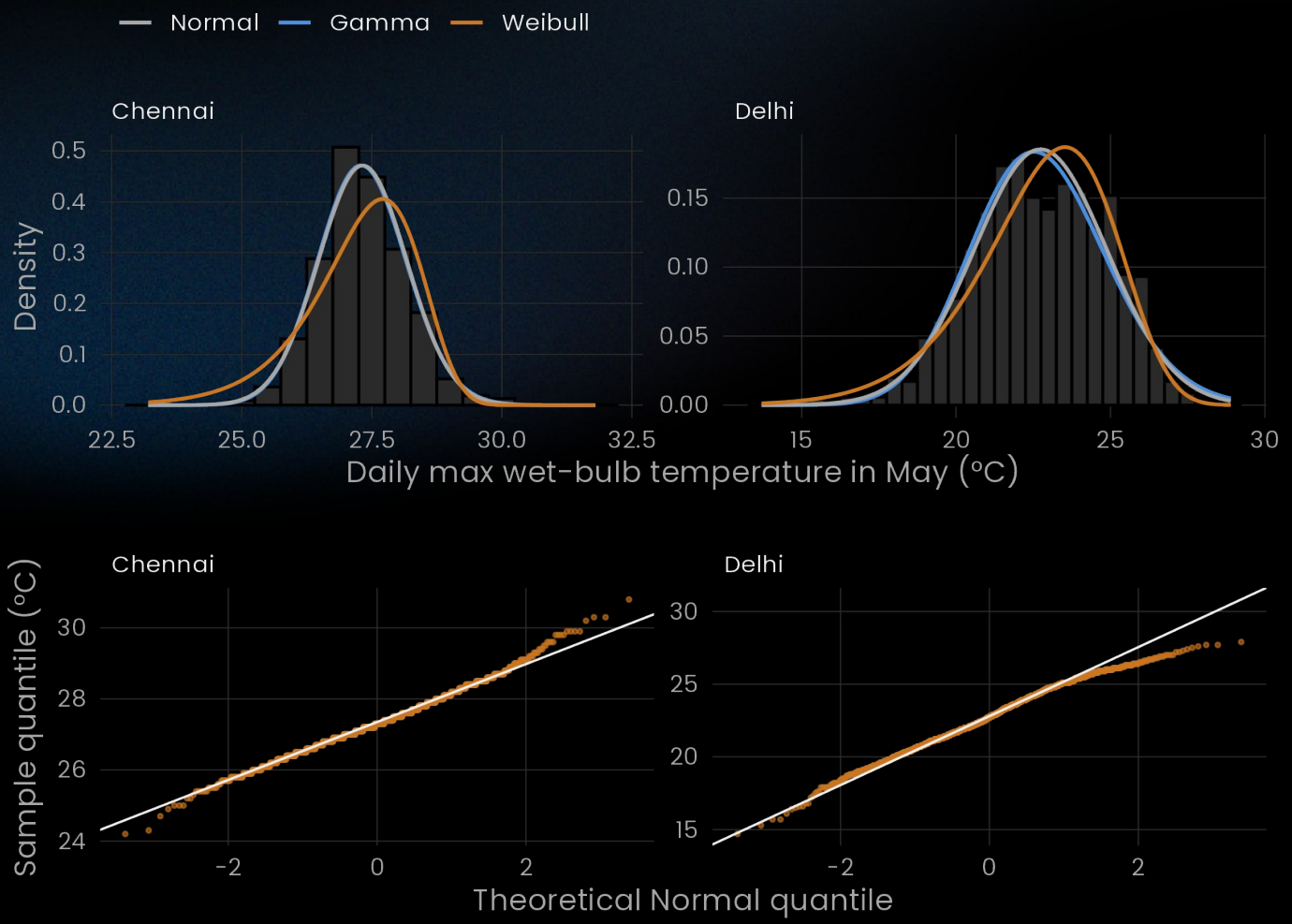
Cell = share of flagged days that were not IMD heatwave days, 1981–2025 (day counts per cell in blind_spot_sensitivity.csv).

TWO HAZARDS, TWO SEASONS, TWO GEOGRAPHIES



Share of days in each calendar week, 1981–2025 (square-root colour scale).

WHICH PROBABILITY MODEL FITS?



Daily wet-bulb (May)

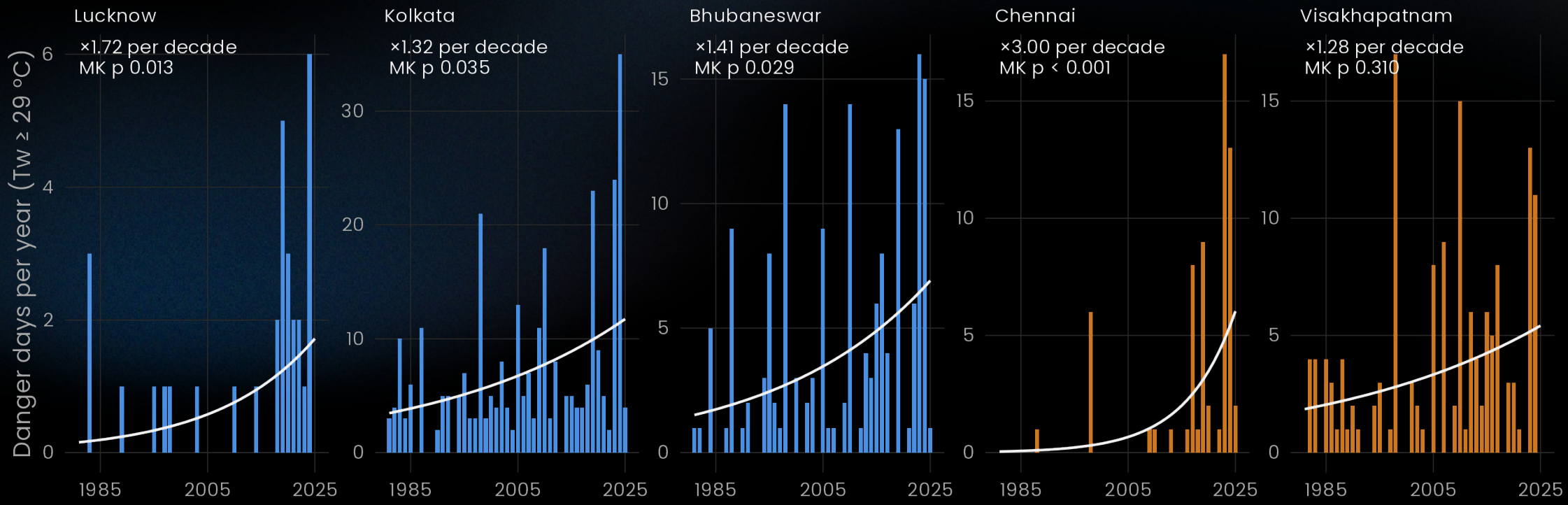
MLE fits of Normal, Gamma, Weibull and Lognormal. Best by AIC: Normal 6, Weibull 2, Lognormal 1, Gamma 1.

Large Anderson–Darling statistics: the bulk fits, the tail does not, so we model the tail with extreme-value theory.

Annual danger-day counts

Variance/mean = 2.0–9.0. Dispersion test rejects Poisson ($p < 0.001$ everywhere); negative binomial wins on AIC and LRT.

RQ2 · IS IT GETTING WORSE?



Cities with ≥ 20 danger days. Line = negative-binomial trend (×IRR per decade); p = Mann–Kendall test (own implementation, tie-corrected).

×1.54 / decade

danger days, NB GLM, 95% CI 1.34–1.77

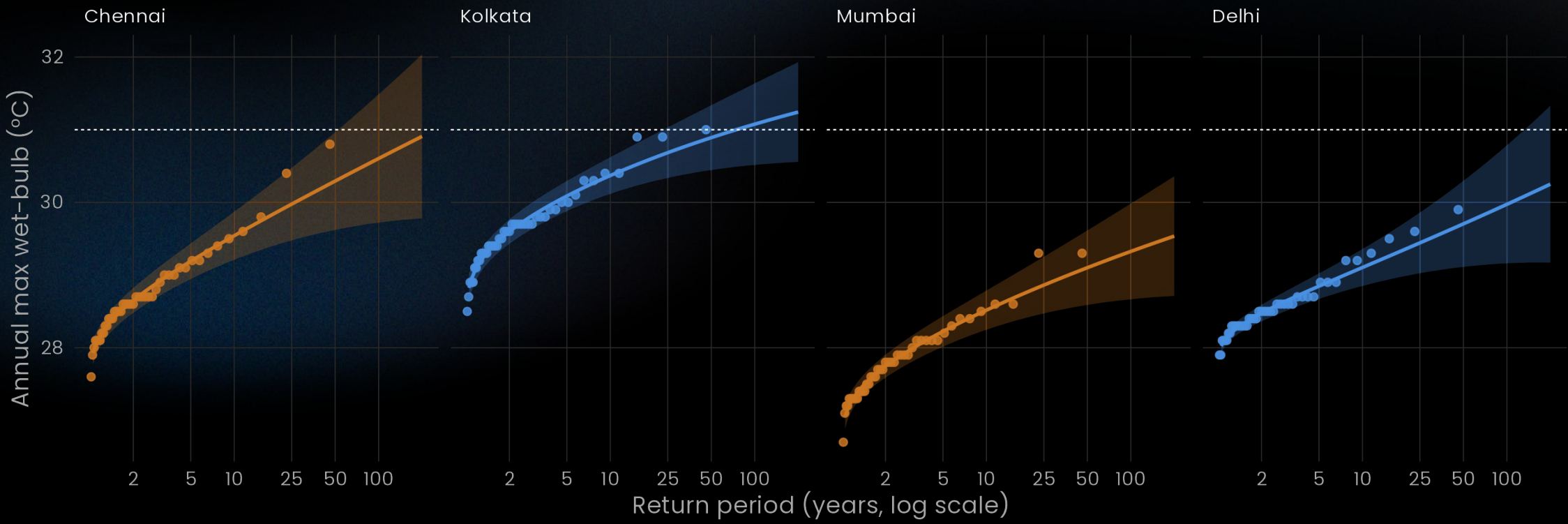
+0.53 °C

Apr–Jun wet-bulb, 2021–25 vs 1981–90 (Tukey; Friedman p < 0.001)

0.35 → 2.85

Chennai danger days/yr, 1981–2000 vs 2006–25 (perm. p = 0.033)

RQ3 · EXTREMES ARE SHIFTING (GEV)



GEV fitted by maximum likelihood (own optim code). Band = 95% delta-method CI; dots = observed annual maxima (Weibull plotting positions). Dashed = 31 °C.

1-in-50 → 1-in-2.6

Chennai: 1985's 50-year humid-heat extreme, under today's climate

7 of 10 cities

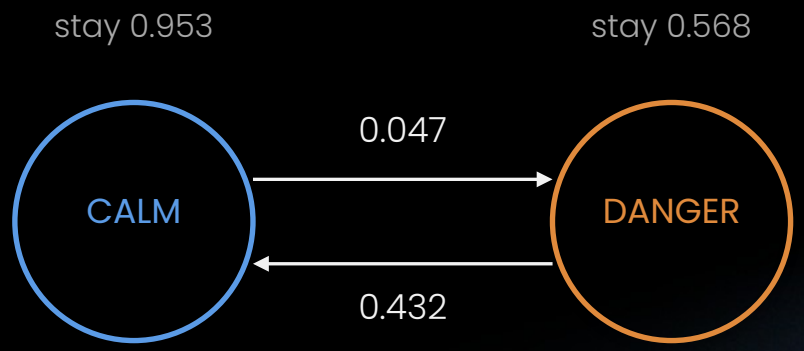
significant upward shift of the GEV location (LRT $p < 0.05$)

35%

chance Kolkata sees a ≥ 31 °C wet-bulb day in 2026–35 (trend GEV)

RQ4 · DANGER COMES IN SPELLS (MARKOV CHAIN)

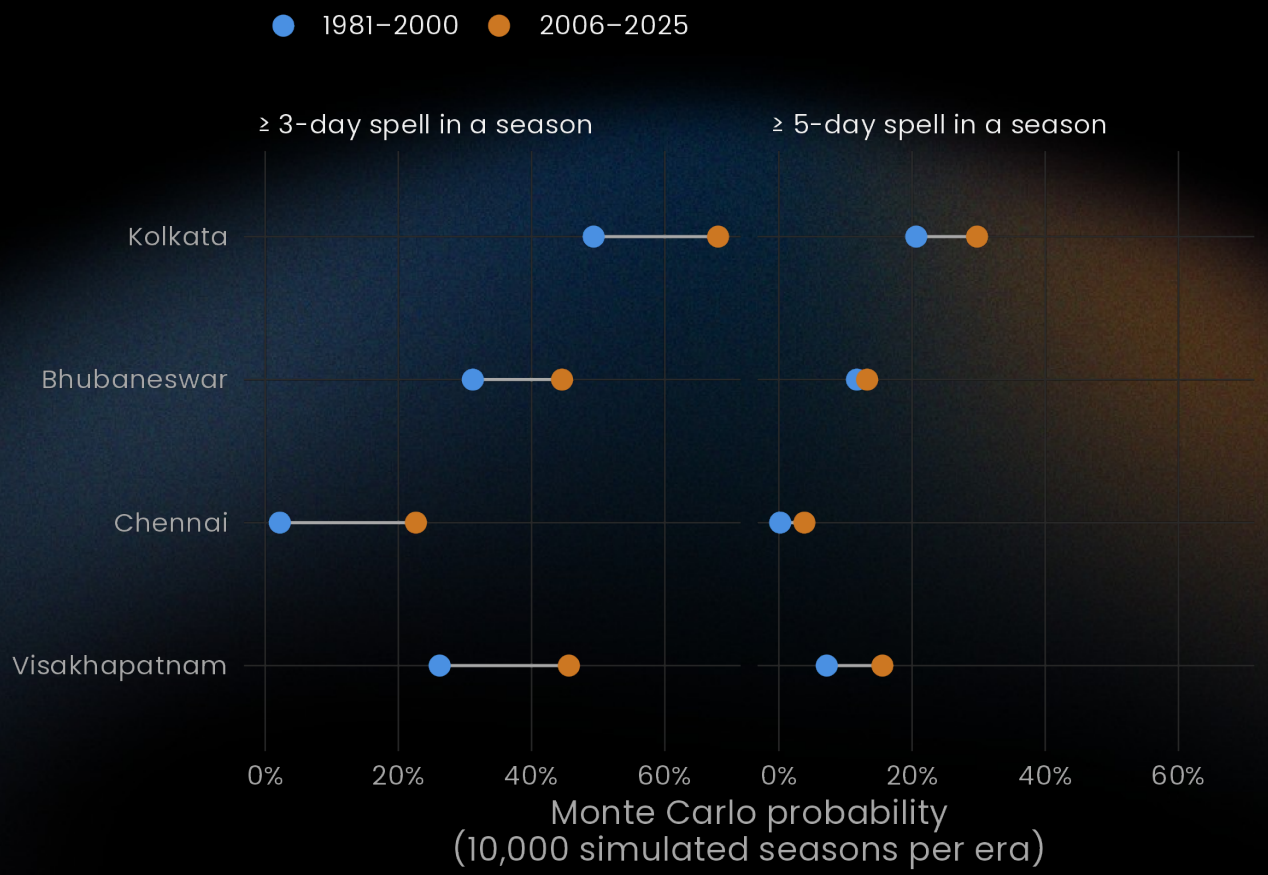
Kolkata, peak season (5 May – 4 Jul)



Independence rejected: χ^2 and G^2 $p < 0.001$; persistence odds ratio 27–41

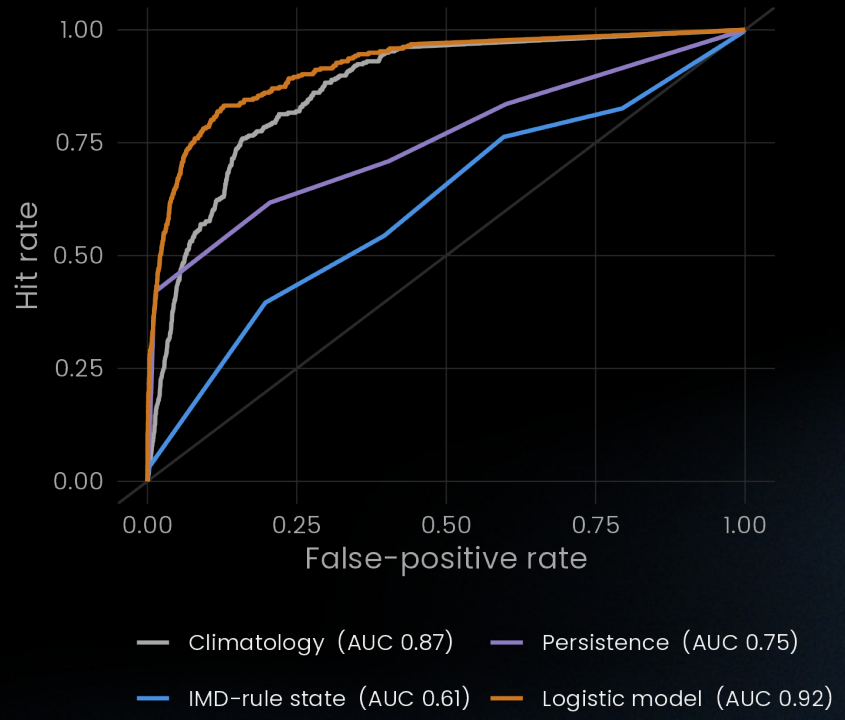
Spells are geometric: mean = $1/(1 - p_{11})$; goodness-of-fit p = 0.10–0.67

Monte Carlo, 10,000 seasons/era: P(≥ 3-day spell) in Chennai 2% → 23%, in Kolkata 49% → 68%

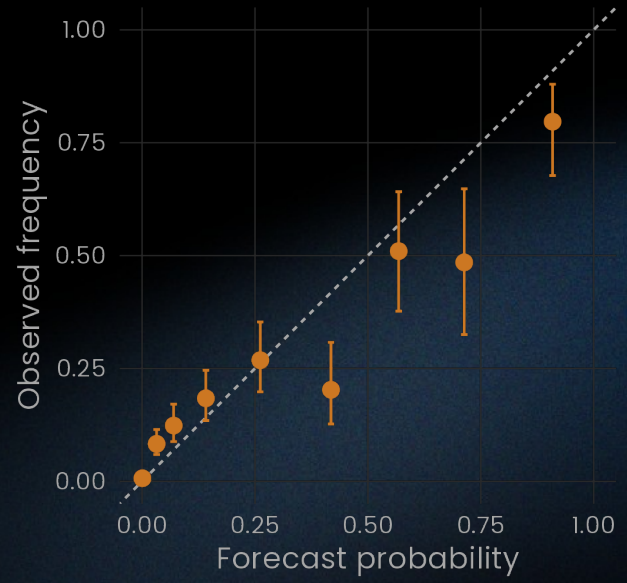


RQ5 · A CALIBRATED NEXT-DAY WARNING

ROC on 2015–2025 test years



Reliability of the logistic model



AUC 0.92

logistic model vs 0.61 for the IMD-rule state (test years 2015–25)

+17%

Brier skill over climatology

4%

of next-day danger days caught by 'IMD heatwave today'

Each +1 °C of today's wet-bulb multiplies tomorrow's odds of danger by 11.3 (95% CI 7.5–17.1).

RQ6 · BASELINE ML, AND WHAT IS NEW

Set-up

Target: will the day +1, +2 or +3 ahead be a danger day (wet-bulb ≥ 29 °C)?

Baselines: climatology, IMD rule, logistic regression, decision tree (rpart), random forest (ranger, 400 trees)

24 features: today's weather, 3–7-day wet-bulb history, season, city

Train 1981–2014 · alert thresholds tuned on 2010–14 · test 2015–25, never seen

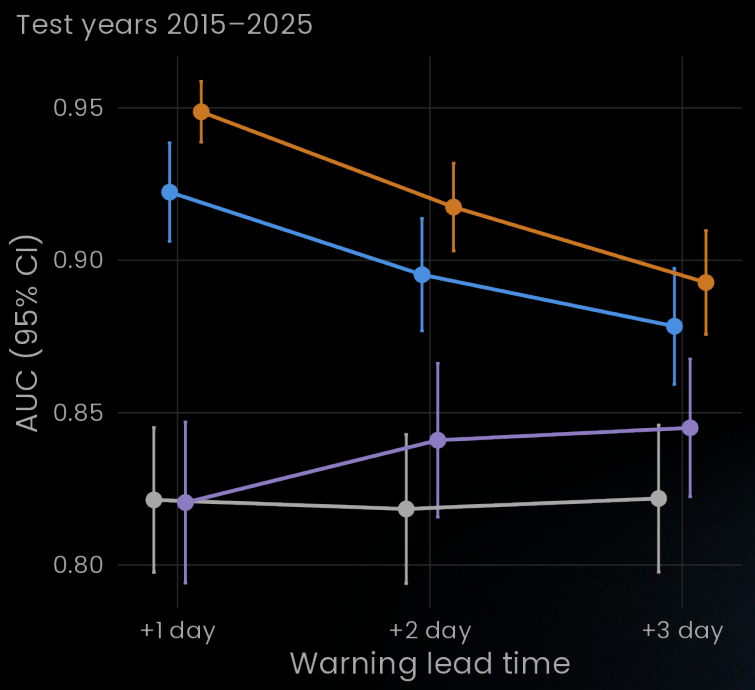
What is new

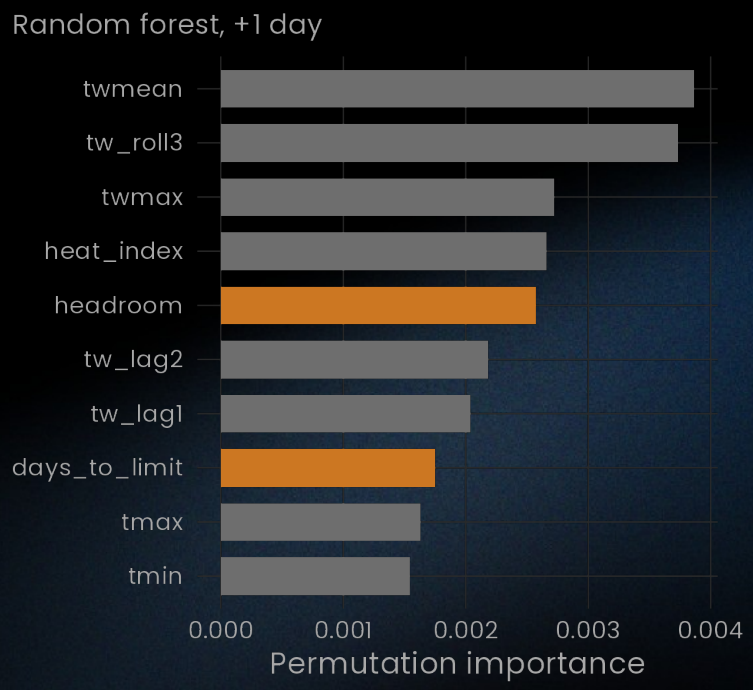
Physiology-informed features
headroom to the 29 °C limit · closing speed · days-to-limit

Hybrid alert, not a replacement
IMD rule OR ML layer, scored on blind-spot recovery

Zero-history cities
leave-one-city-out: warn where no labelled past exists

RQ6 · RANDOM FOREST LEADS AT EVERY LEAD TIME





Climatology · Logistic · Decision tree · Random forest · Weather · Physiology-informed

AUC 0.95

random forest, +1 day (logistic 0.92, IMD rule 0.61)

AUC 0.89

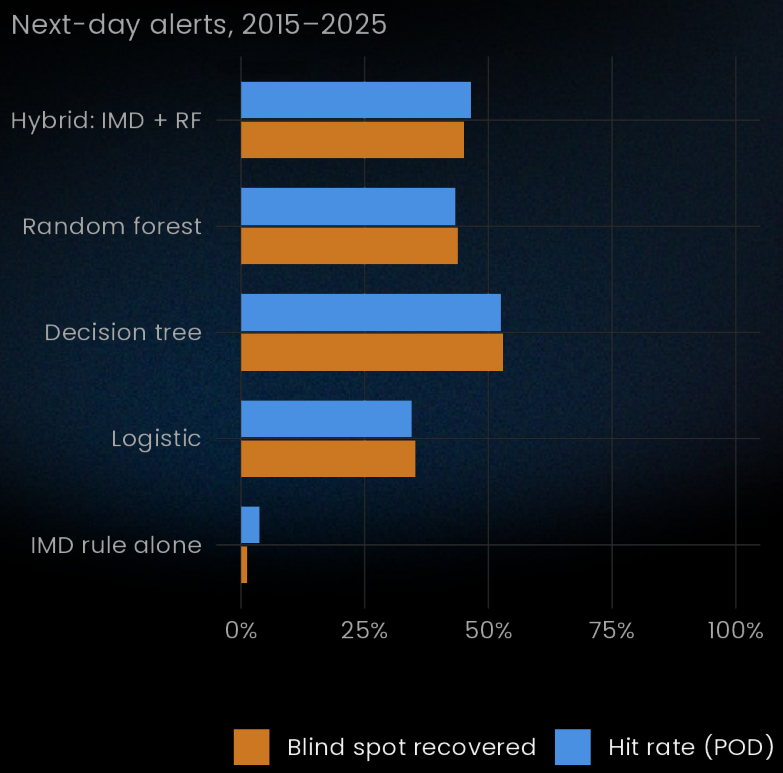
still at +3 days, vs 0.82 for climatology

+22%

Brier skill over climatology, +1 day

Headroom to the physiological limit ranks #5 of 24 features. One tree is the weakest model; a forest of 400 fixes its variance.

NOVELTY · CLOSING THE BLIND SPOT, ANYWHERE





45%

of the blind spot recovered by the hybrid alert (IMD rule alone: 1%)

47%

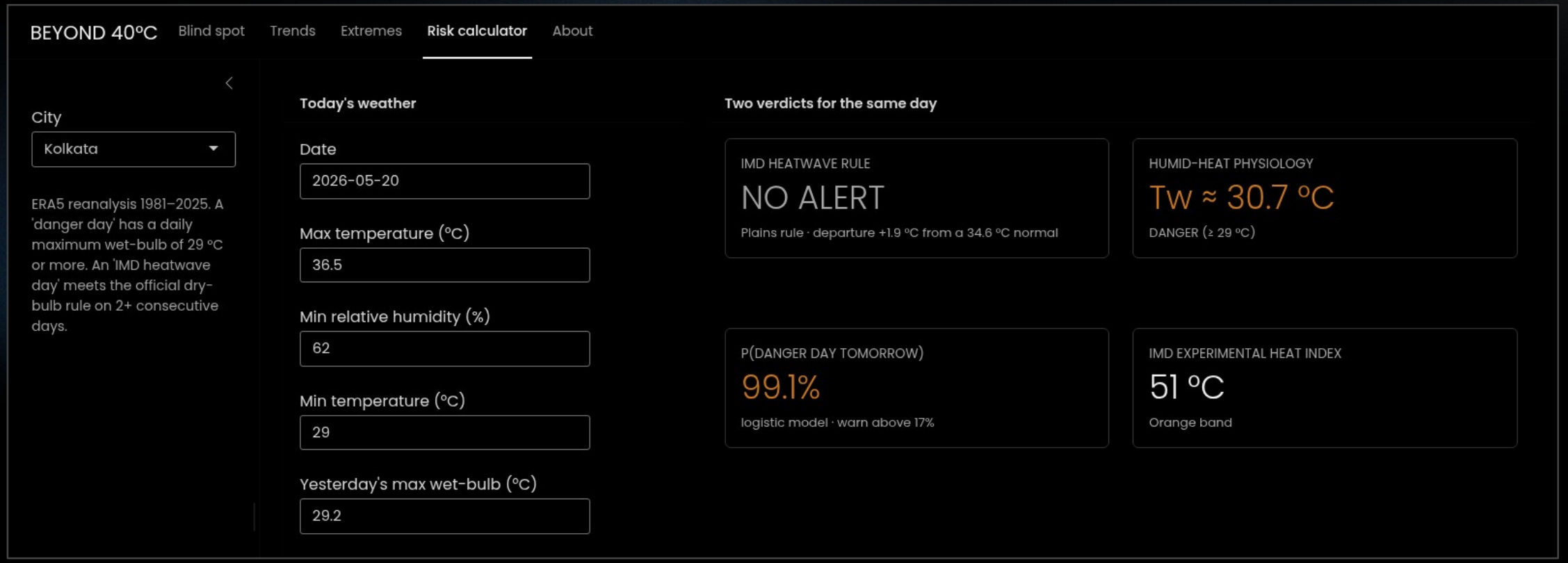
of next-day danger days caught (IMD rule: 4%)

0.93 / 0.94

mean AUC in an unseen city / a trained city

Honest cost: 64% of hybrid alerts are false alarms. That suits a cheap hydrate-and-rest advisory, not school closures.

SHINY DASHBOARD: THE GAP IN ONE SCREEN



Risk calculator, Kolkata on 20 May: the IMD rule says NO ALERT, the wet-bulb (30.7 °C) says DANGER, and the model gives 99% for tomorrow.

Rscript -e 'shiny::runApp("app")' · 5 tabs: blind spot, trends, extremes, risk calculator, about

CONCLUSIONS & RECOMMENDATIONS

RQ1 The IMD rule misses 96% of dangerous humid-heat days, in every city

RQ2 Danger days grow ×1.54 per decade; Apr–Jun wet-bulb is up 0.53 °C

RQ3 Tail shifting in 7/10 cities; Chennai's 1985 50-year extreme is now 1-in-2.6

RQ4 Danger persists in spells; 3-day spells are now far more likely

RQ5 A logistic warning reaches AUC 0.92 vs 0.61, and is calibrated

RQ6 A random forest reaches AUC 0.95, recovers 45% of the blind spot, and transfers to unseen cities

- ↗ Add a humidity-aware trigger (wet-bulb or Heat Index) to heatwave declarations
- ↗ Replace fixed national cut-offs with local, probabilistic thresholds and return periods
- ↗ Prioritise Kolkata, Bhubaneswar, Visakhapatnam and Chennai, where silent days dominate

LIMITATIONS & FUTURE WORK

Limitations

ERA5's 0.25° grid smooths peaks: only 2 grid-days reached 31 °C; Mumbai's cell is sea-moderated

One grid cell stands in for IMD's two stations, which flatters IMD, so our miss rates are lower bounds

Thresholds come from young, healthy adults; older people and outdoor workers are at risk sooner

Day-level tests are approximate under autocorrelation (mitigated with block bootstrap and Markov models)

Future work

Repeat with IMD station observations and finer reanalysis (ERA5-Land, IMDAA)

Link to hospital or mortality records to calibrate thresholds by age and occupation

Extend to warm nights and multi-day cumulative heat load

Operational district-level forecasts using weather-model inputs instead of today's observations

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Mann (1945); Sen (1968); DeLong et al. (1988), comparing correlated ROC curves.

THANK YOU